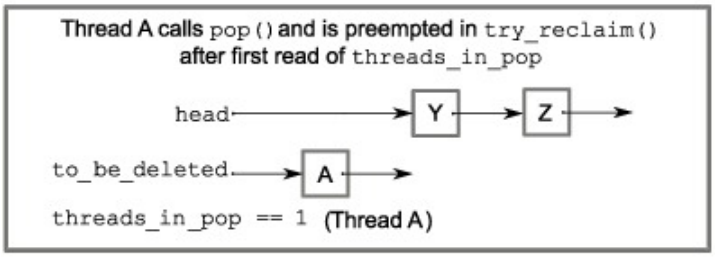
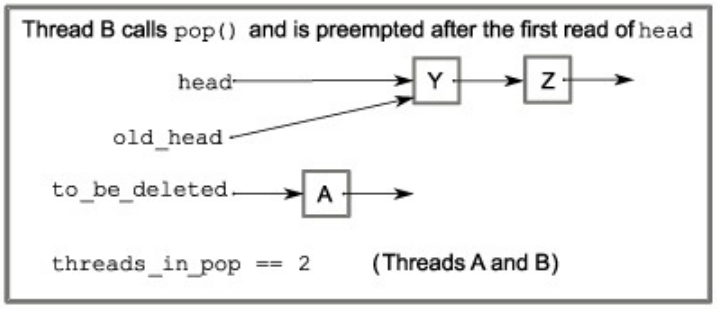
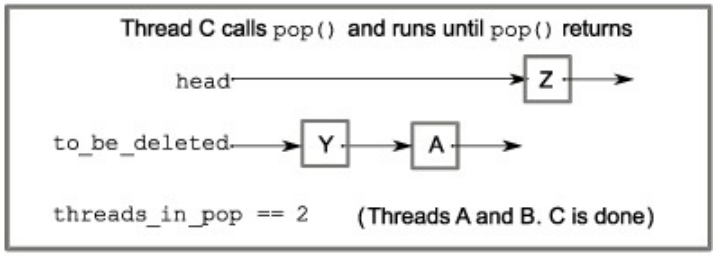
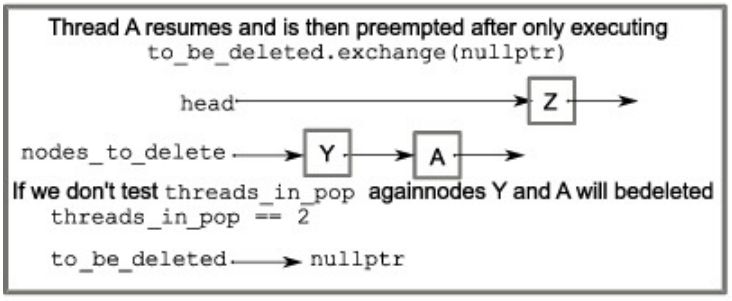
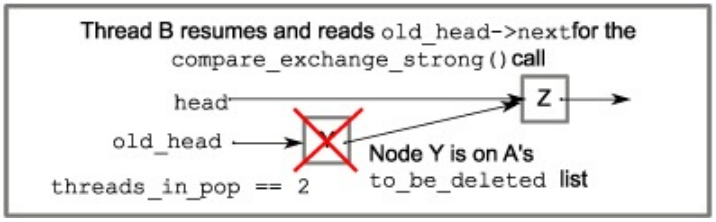

Initially
head
X
Y
Z
to_be_deleted
A
threads_in_pop == 0


Thread A calls pop() and is preempted in try_reclaim() after first read of threads_in_pop
head
Y
Z
to_be_deleted
A
threads_in_pop == 1 (Thread A)


Thread B calls pop() and is preempted after the first read of head
head
Y
Z
old_head
to_be_deleted
A
threads_in_pop == 2 (Threads A and B)


Thread C calls pop() and runs until pop() returns
head
Z
to_be_deleted
Y
A
threads_in_pop == 2 (Threads A and B. C is done)


Thread A resumes and is then preempted after only executing to_be_deleted.exchange(nullptr)
head
Z
nodes_to_delete
Y
A
If we don't test threads_in_pop againnodes Y and A will bedeleted
threads_in_pop == 2
to_be_deleted
nullptr


Thread B resumes and reads old_head->next for the compare_exchange_strong() call
head
Z
old_head
Node Y is on A's to_be_deleted list
threads_in_pop == 2